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Difference-in-Differences Estimator of Quantile Treatment Effect on the Treated

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ABSTRACT

We propose a new difference-in-differences (DID) estimator of the quantile treatment effect on the treated (QTT). The model assumes a common time effect on the cumulative distribution functions of untreated potential outcomes, allowing for covariates. This condition holds if and only if the net change in the untreated outcome densities is common across treated and control groups. Unlike the Changes-in-Changes model our model is compatible with the usual DID assumption for means, and it provides a computationally simple and straightforward way to control for covariates. We establish uniform consistency and weak convergence of the proposed estimator of QTT and the related functions. The estimators and the simultaneous confidence bands remain valid even for discrete outcome variables. As an empirical application, the distributional impact of the earned income tax credit on birth weight is investigated. We provide a STATA ado file package.

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1. Introduction

In this article, we propose a difference-in-differences estimator of the quantile treatment effect on the treated (QTT). The setup is the canonical quasi-experimental design with repeated cross-section data. Use of panel data is not necessary. Observed dataset contains a response variable y_i and individual characteristics $\mathbf{x}_i \in \mathbb{R}^{d_x}$, where i indexes observation. There are two groups of individuals, treated ($d_i = 1$) and untreated ($d_i = 0$), that are observed at two points in time, pre-treatment period ($t_i = 0$) and post-treatment period ($t_i = 1$). Only the treated group receives the treatment between the two periods. Denote the potential outcome of y (hypothetically) without the treatment as y^0 , and with the treatment as y^1 where the superscript denotes the treatment status. Then, the object of interest is the QTT at the post-treatment period

$$F_{y_i}^{\leftarrow}(\tau | d_i = 1, t_i = 1) - F_{y_i}^{\leftarrow}(\tau | d_i = 1, t_i = 0) \quad (1)$$

for $\tau \in (0, 1)$, where $F_y^{\leftarrow}(a|x)$ denotes the conditional a -quantile of y given x . The first term in (1) is observable while the second term is the counterfactual quantile function (QF) in the absence of the treatment. The credibility of the final estimation and inference results essentially depends on how we model the counterfactual outcome distribution for the treated group.

For identification of QTT, we assume common time effects on the (cumulative) distribution functions (DF) of the untreated potential outcomes y^0 . The same condition (up to conditioning) has been studied by Roth and Sant'Anna (2023, Proposition 3.1) in characterizing the transformation-invariant parallel trends. It is called the distributional difference-in-differences

(DDID) assumption in this article. Since we allow covariates as conditioning variables, observationally different subpopulations may have unparallel temporal changes in the distribution of y^0 . To properly guide its empirical plausibility, by revisiting Roth and Sant'Anna (2023, Proposition 3.2), the assumption is formally characterized as the *net changes* of the untreated outcome density being common across treated and control groups. While Roth and Sant'Anna consider the same identifying assumption as in this article, they do not propose QTT estimator under this assumption. To estimate the counterfactual quantile function, we consider estimating the counterfactual distribution function (DF) by a simple two-stage procedure, and taking its (left-)inverse. The two-stage procedure enables controlling for covariates via straightforward computation with standard statistical software packages. Under standard assumptions for continuous outcome y , we show that the proposed QTT estimator is uniformly consistent, and weakly converges to a Gaussian process achieving semiparametric efficiency. To make inference valid for generic (i.e., noncontinuous) outcome y , we use the confidence-band-inverting method of Chernozhukov et al. (2020; CFMW). The confidence bands for the status quo and counterfactual DFs are constructed by bootstrap, and those for QFs and QTT are computed by left-inverting the DF-bands, and taking the Minkowski difference of the inverted DF-bands, respectively. Even for generic outcome variables, the proposed estimator and confidence bands are still valid. For instance, when the outcome has a discrete distribution with a finite support or a bounded mixture distribution with a finite number of measure-zero intervals in $\mathbb{R}$, the QTT estimator is almost uniform consistent and the confidence bands simultaneously cover QFs and QTT with a prespecified probability.

Related Literature

The key idea of natural experiments (or DID models) is to assume that the effect of confounding factors that changed over time is common for the treated and untreated groups. The so-called “common time effect” or “common trend” assumption is typically imposed on some feature (e.g., mean) of the untreated potential outcome distributions. For estimating the average treatment effect on the treated (ATT), the approach has been widely applied in numerous empirical studies on program evaluations following the seminal work by Ashenfelter (1978) and Ashenfelter and Card (1985). Some literature reviews on DID include Meyer (1995), Lechner (2011), Deschenes and Meng (2018), de Chaisemartin and D’Haultfoeuille (2022), and Roth et al. (2022). Recently, Abadie (2005) extended the method to allow heterogeneous time and treatment effect nonparametrically by implementing the idea of conditioning (Heckman, Ichimura, and Todd 1997, 1998). He imposes the common time effect assumption on the conditional average of untreated potential outcomes.

Our article bridges a gap in the literature by relating Abadie’s (2005) model to DID estimation of QTT. The literature on DID estimation of QTT was pioneered by Athey and Imbens (2006; AI). AI proposes the changes-in-changes (CIC) model which essentially assumes a common time effect in terms of a “shift function” (Doksum 1974). (See Section 2.2 for details.) Melly and Santangelo (2015, manuscript) extend AI (2006) to allow for conditioning covariates. Callaway, Li, and Oka (2018) and Callaway and Li (2019) consider panel data DID methods assuming that the conditional distribution of the individual-level time effect on untreated outcomes is common. Bonhomme and Sauder (2011) impose a common time effect on the logarithm of the characteristic function. Compared to these existing methods, particularly the CIC estimator, our QTT estimator is different in four important aspects, the first two of which are implications of Roth and Sant’Anna (2023). First, the DDID assumption is compatible with the usual DID assumption for means regardless of the functional form. So, under DDID, the usual DID estimator of ATT can be used without worrying about scaling. In contrast, the CIC model identifies ATT differently and results in a different estimator. Second, the new DID assumption implies a testable shape restriction. This property is particularly desirable for QTT estimation since identification of QTT typically requires a much stronger model assumption compared to the case of ATT. Our method is the only one proposed so far that accompanies such a testable restriction. Third, the uniform inference procedure is well-established for generic outcome variables. In particular, outcome variables can have discrete/mixture distributions, which provides a much wider applicability compared to other methods. Fourth, controlling for covariates is achieved by a computationally simple and straightforward two-stage procedure. This is another great advantage over the CIC estimator which typically requires complex computational steps to control for covariates.

The rest of the article is organized as follows. In Section 2, the model assumption and the identification result are presented and discussed. Section 3 covers estimation and inference. Section 4 explains the .ado file package for STATA and contains simulation studies. Section 5 contains concluding remarks. The

empirical application can be found in the online supplementary material.

2. Model

This section presents the identification result. The key model assumption, distributional DID, is compared to the Changes-in-Changes (CIC) model, and it is characterized by a useful decomposition result by revisiting Roth and Sant’Anna (2023). Extension to the case of multilevel treatment is also discussed.

2.1. Identification

Identification of DFs and QFs is established. The key identifying condition is the distributional DID (DDID) assumption. Let $\mathcal{Y}$ be a closed interval in the extended real line $\overline{\mathbb{R}}$. Unless specifically mentioned, the region of interest $\mathcal{Y}$ is set to be $\overline{\mathbb{R}}$. Define $F_{y_i^j|\mathbf{x}_i,dt}(y) \equiv P(y_i^j \leq y | \mathbf{x}_i = \mathbf{x}, d_i = d, t_i = t)$, and $F_{y_i^j|dt}(y)$ similarly for $j = 0, 1$.

Assumption 1 (Distributional DID). For all $y \in \mathbb{R}$ and $\mathbf{x} \in \text{spt}(\mathbf{x}_i)$,

$$F_{y_i^0|\mathbf{x}_i,11}(y) - F_{y_i^0|\mathbf{x}_i,10}(y) = F_{y_i^0|\mathbf{x}_i,01}(y) - F_{y_i^0|\mathbf{x}_i,00}(y) \quad (2)$$

Assumption 1 requires in the absence of the treatment, the proportion of treatment group units with a response below y would have changed similarly from period 0 to 1 in the treatment and in the control group, for all y . It turns out that Assumption 1 is just another convenient way to express common time effects on the probability density (or mass) of each point y . (See Section 2.3 for details.) Assumption 1 allows observationally different subpopulations of each group to have unparallel temporal changes in distributions of y_i^0 .¹ It also does not restrict QTT to be the same across different subpopulations. The condition is invariant to monotone transformation (or scale) in the sense that if Assumption 1 holds for y_i^0 , then it holds for any strictly increasing transformation of y_i^0 . Moreover, it implies the usual DID assumption for means (RS, 2023; Proposition 3.1). In contrast, CIC implies a different model for the mean (AI 2006; Expression (16) on p. 441). Thus, unlike CIC, Assumption 1 does not contradict the usual DID assumption. In this article, we also allow the distribution of $(d_i, \mathbf{x}_i)$ to be time-varying so that the resulting estimator becomes robust to compositional changes that occur without violating Assumption 1. If $(d_i, \mathbf{x}_i)$ is independent from t_i , then we can impose $P(d_i = d, t_i = t | \mathbf{x}_i) = P(d_i = d | \mathbf{x}_i) P(t_i = t)$ in the rest of this article. The results will remain valid with trivial modifications.

We additionally impose two assumptions. Define $p_{dt}(\mathbf{x}) \equiv P(d_i = d, t_i = t | \mathbf{x}_i = \mathbf{x})$ and $p_{dt} \equiv P(d_i = d, t_i = t)$.

Assumption 2 (Common Support). There exists $c > 0$ such that $c < p_{dt}(\mathbf{x}) < 1 - c, \forall \mathbf{x} \in \text{spt}(\mathbf{x}_i)$.

¹We also implicitly assume that the treated group would have the same pre-period distribution of the untreated potential outcome regardless of the treatment history.

Assumption 3 (Repeated Cross-sections). $(y_i, \mathbf{x}_i, d_i)$ is iid conditional on t_i .

Assumption 2 requires that each subpopulation, characterized by observed $\mathbf{x}$, includes both treated and untreated individuals. It rules out irregular identification (Khan and Tamer 2010). If observations with low $p_{dt}(\mathbf{x})$ are trimmed, one may consider applying the bias correction method of Sasaki and Ura (2022). **Assumption 3** produces a mixture distribution that is less restrictive compared to the one studied in Abadie (2005), as we do not assume that t_i affects the joint distribution of $(y_i, \mathbf{x}_i, d_i)$ only through (y_i^0, y_i^1) . Effectively, $(y_i, \mathbf{x}_i, d_i, t_i)$ is assumed to be iid following the mixture distribution

$$\begin{aligned} P_M(y_i \leq y, \mathbf{x}_i \leq \mathbf{x}, d_i = d, t_i = t) \\ = \lambda t P(y_i \leq y, \mathbf{x}_i \leq \mathbf{x}, d_i = d | t_i = 1) \\ + (1 - \lambda)(1 - t) P(y_i \leq y, \mathbf{x}_i \leq \mathbf{x}, d_i = d | t_i = 0) \end{aligned}$$

where $\lambda = P(t_i = 1)$. Note that conditioning on t_i is kept here.

Prior to presenting the identification result for status quo/counterfactual DFs and QFs, we provide a formal definition of the left-inverse operator below. It is slightly different from the usual one since its value is restricted to be in $\mathcal{Y}$. The definition is from CFMW (2020).

Definition 1 (Left-inverse). Given a monotone increasing function $y \mapsto G(y)$, the left inverse of G is $G^\leftarrow(a) \equiv \inf\{y \in \mathcal{Y} : G(y) \geq a\}$ if $\sup_{y \in \mathcal{Y}} G(y) \geq a$ and $G^\leftarrow(a) \equiv \sup\{y \in \mathcal{Y} : G(y) \leq a\}$ otherwise.

Theorem 1 (Identification). Suppose **Assumptions 1–3**. Then, $F_{y_i^1|11}(y)$, $F_{y_i^0|11}(y)$, $F_{y_i^1|11}^\leftarrow(\tau)$ and $F_{y_i^0|11}^\leftarrow(\tau)$ are identified for $y \in \mathbb{R}$ and $\tau \in (0, 1)$. In particular, we have

$$\begin{aligned} F_{y_i^0|11}(y) = E[(p_{11}(\mathbf{x}_i)/p_{11})\{F_{y_i|\mathbf{x}_i,10}(y) \\ + F_{y_i|\mathbf{x}_i,01}(y) - F_{y_i|\mathbf{x}_i,00}(y)\}]. \end{aligned} \quad (3)$$

Proof. To show identification of $F_{y_i^0|11}(y)$ for $y \in \mathbb{R}$, as in the proof of Lemma 1 of Firpo (2007),

$$\begin{aligned} F_{y_i^0|11}(y) &= P(y_i^0 \leq y, d_i t_i = 1) / p_{11} \\ &= E[P(y_i^0 \leq y, d_i t_i = 1 | \mathbf{x}_i) / p_{11}] \\ &= E[(p_{11}(\mathbf{x}_i) / p_{11}) F_{y_i^0|\mathbf{x}_i,11}(y)] \\ &= E[(p_{11}(\mathbf{x}_i) / p_{11})\{F_{y_i^0|\mathbf{x}_i,10}(y) \\ &\quad + F_{y_i^0|\mathbf{x}_i,01}(y) - F_{y_i^0|\mathbf{x}_i,00}(y)\}] \end{aligned}$$

where division by zero is avoided by **Assumption 2** and all population quantities are with respect to the mixture distribution generated by **Assumption 3**. From the last expression, the result follows. Now, to show identification of $F_{y_i^0|11}^\leftarrow(\tau)$ for $\tau \in (0, 1)$, it suffices to prove that the left-inverse of $F_{y_i^0|11}(y)$ is well-defined for each τ . Suppose $\tau \in (0, 1)$. Since $F_{y_i^0|11}(y)$ is monotone increasing in y , $F_{y_i^0|11}(-\infty) = 0$, and $F_{y_i^0|11}(+\infty) = 1$, the set $\{y \in \mathcal{Y} : F_{y_i^0|11}(y) \geq \tau\}$ is nonempty and bounded below by an element in $\mathbb{R}$. Hence, the result. Proofs for $F_{y_i^1|11}(y)$ and $F_{y_i^1|11}^\leftarrow(\tau)$ are omitted. $\square$

Based on (3), we can derive a moment condition with inverse probability weighting, which motivates the two-stage estimator proposed in Section 3.1. Alternatively, the distribution regression of Chernozhukov, Fernández-Val, and Melly (2013; CFM) could be used to estimate conditional DFs in (3). The proposed estimator has a computational advantage (e.g., in bootstrap) since it does not require the estimation of conditional DFs (at each grid point). When the computational cost is affordable, another potentially interesting path would be doubly-robust type estimation (e.g., Wooldridge (2010, Chapter 21); Sant’Anna and Zhao 2020) which is known to enjoy nice robustness properties.

Another implication of (3) is the monotonicity of the right-hand side expression of (3) in y . It follows immediately from the fact that $F_{y_i^0|11}(y)$ is a cumulative distribution function. Without **Assumption 1**, the monotonicity may not hold in general since the limit of $F_{y_i^0|\mathbf{x},01}(y) - F_{y_i^0|\mathbf{x},00}(y)$ is zero as y tends to $+\infty$ or $-\infty$ (i.e., the difference is either constantly zero or non-monotone in y). **Corollary 1** summarizes the result. This property can be used to test **Assumption 1**. See Section 3.4 for details.

Corollary 1. Under **Assumptions 1–3**, the right-hand side of (3) is monotone increasing in y .

2.2. Comparison with Changes-in-Changes Model

The advantages of the proposed model are discussed in comparison with the Changes-in-Changes model of AI (2006). To make the comparison more informative, we first aim to state each key assumption in the form of a “common time effect” on some feature of the error distribution. In Section 2.2.1, we show that a DDID assumption imposed on the error term is sufficient for our key identifying assumption (**Assumption 1**). In Section 2.2.2, we show that common rank-preserving effect on the error is sufficient for the CIC identification result. The condition is weaker than the within-group time invariance assumption considered by AI (2006). Then, comparing these two sets of assumptions, we discuss the benefits of using our method.

2.2.1. An Example of Structural Assumptions for DDID

Two assumptions are made for the structural function that generates the untreated potential outcome and the conditional distribution of unobservables. Define $F_{u_i|\mathbf{x}_i,dt}(u)$ similarly to $F_{y_i^j|\mathbf{x}_i,dt}(y)$.

Assumption 4 (Structural Equation). $y_i^0 = h(\mathbf{x}_i, u_i)$ where h is Borel measurable and the unobservable $u_i \in \mathbb{R}^{d_u}$ has no element being a deterministic function of d_i or t_i .

In **Assumption 4**, the structural function h depends on covariates $\mathbf{x}$ and unobservable u but not directly on time as in AI (2006). While the structural relationship that produces y^0 is time-invariant, time can affect the outcome y^0 via $\mathbf{x}$ or u . We assume no element of u is a deterministic function of d_i or t_i . The structural function for y_i generally depends on the treatment status $w_i \equiv d_i \times t_i$ but here we define

$y_i^0 \equiv h(0, \mathbf{x}_i, u_i) = h(\mathbf{x}_i, u_i)$ for simplicity. We do not require h to be monotonic in u .

Assumption 5 (Distributional DID for Unobservables). For all $u \in \mathbb{R}^{d_u}$ and $\mathbf{x} \in \text{spt}(\mathbf{x}_i)$,

$$F_{u_i|\mathbf{x},11}(u) - F_{u_i|\mathbf{x},10}(u) = F_{u_i|\mathbf{x},01}(u) - F_{u_i|\mathbf{x},00}(u) \quad (4)$$

Assumption 5 is similar to Assumption 1 except that it is imposed on the unobservable u_i . It states that conditional on $\mathbf{x}$, both groups experienced the same change in the probability distribution of unobserved variables u_i over time. The distribution of u_i may vary across periods and groups. In particular, u_i is not assumed to be independent of time conditional on each group. Then, Assumption 1 is implied by Assumptions 4 and 5 since

$$\begin{aligned} F_{y_i^0|\mathbf{x}_i,11}(y) &= \int 1_{(h(\mathbf{x}_i,u) \leq y)} dF_{u_i|\mathbf{x}_i,11}(u) \\ &= \int 1_{(h(\mathbf{x}_i,u) \leq y)} dF_{u_i|\mathbf{x}_i,10}(u) + \int 1_{(h(\mathbf{x}_i,u) \leq y)} dF_{u_i|\mathbf{x}_i,01}(u) \\ &\quad - \int 1_{(h(\mathbf{x}_i,u) \leq y)} dF_{u_i|\mathbf{x}_i,00}(u) \end{aligned}$$

2.2.2. A Weaker Identifying Condition for CIC

A weaker identifying restriction for the CIC model is presented. Since the identification result of AI (2006; Theorem 3.1) assumes that the error term is a scalar and does not allow covariates but the time dummy, we consider a restricted model by assuming u_i to be a scalar and by dropping covariates $\mathbf{x}_i$ in Assumption 4. We add t_i as an additional argument of h to allow direct dependence of h on t . Denote the inverse of h given t as $h^{-1}(\cdot; t)$. Assume further that h is strictly monotonic in u_i as in AI (2006). $\mathbb{U}_{dt}$ and $\mathbb{Y}_{dt}$ denote the support of $u_i|dt$ and $y_i^0|dt$, respectively. Assume $\mathbb{U}_{1t} \subseteq \mathbb{U}_{0t}$ and $\mathbb{Y}_{1t} \subseteq \mathbb{Y}_{0t}$ for $t = 0, 1$. Then, we can show that the following condition is essential for the identification result of AI (2006): for almost every $u \in \mathbb{U}_{11}$, there exists $\Delta_U(u)$ such that

$$\begin{aligned} F_{u_i|11}(u) &= F_{u_i|10}(u + \Delta_U(u)) \text{ and} \\ F_{u_i|01}(u) &= F_{u_i|00}(u + \Delta_U(u)) \end{aligned} \quad (5)$$

where $u + \Delta_U(u)$ is either a probability mass point or an interior point of the support of $F_{u_i|00}$.

The sufficiency of (5) follows² from Proposition 1 when h is strictly monotonic in u .

Proposition 1 (Changes-in-Changes). Suppose that the distribution of y_i^0 is either discrete or (absolutely) continuous. Assume that for almost every $y \in \mathbb{Y}_{11}$, there exists $y + \Delta(y) \in \mathbb{Y}_{10}$ such that (i) $F_{y_i^0|11}(y) = F_{y_i^0|10}(y + \Delta(y))$ and $F_{y_i^0|01}(y) = F_{y_i^0|00}(y + \Delta(y))$, and (ii) $y + \Delta(y)$ is either a probability mass point or an interior point of the support of $F_{y_i^0|00}$. Then, we have $F_{y_i^0|11}(y) = F_{y_i^0|10}(F_{y_i^0|00}^{\leftarrow}(F_{y_i^0|01}(y)))$ for almost every $y \in \mathbb{Y}_{11}$.

Proof. From the second equation of (i), we have $F_{y_i^0|00}^{\leftarrow}(F_{y_i^0|01}(y)) = F_{y_i^0|00}^{\leftarrow}(F_{y_i^0|00}(y + \Delta(y))) = y + \Delta(y)$. The

result follows by substituting $y + \Delta(y)$ in the first equation of (i). $\square$

Remark 1. The model assumptions of AI (2006) imply the condition (i). Note $F_{u_i|dt}(u)$ does not depend on t . As shown in the proof of Theorem 3.1 in AI (2006), regarding the first equation of (i), we have $F_{y_i^0|11}(y) = F_{u_i|11}(h^{-1}(y; 1)) = F_{u_i|10}(h^{-1}(y; 1)) = F_{y_i^0|10}(h(h^{-1}(y; 1), 0))$. For the second equation of (i), in $F_{y_i^0|00}(h(u, 0)) = F_{u_i|00}(h^{-1}(h(u, 0); 0)) = F_{u_i|00}(u)$, substituting $u = F_{u_i|00}^{-1}(F_{y_i^0|01}(y)) = F_{u_i|01}^{-1}(F_{y_i^0|01}(y)) = h^{-1}(y; 1)$, we get $F_{y_i^0|00}(h(h^{-1}(y; 1), 0)) = F_{u_i|00}(F_{u_i|00}^{-1}(F_{y_i^0|01}(y))) = F_{y_i^0|01}(y)$. Thus, we can take $\Delta(y) = h(h^{-1}(y; 1), 0) - y$.

Remark 2. Condition (ii) is not implied by AI (2006) model. However, the argument based on applying " $F_{y_i^0|00}^{-1}$ " in their proof of Theorem 3.1 requires a refined condition. First note that for a generic cdf F , we have $F^{\leftarrow}(F(x)) \leq x$ for $x \in \mathbb{R}$. If x is either a probability mass point or an interior point of the support, then it holds as an equality $F^{\leftarrow}(F(x)) = x$. However, x being a point in the support is not sufficient. (A counter-example: X follows uniform distribution on $[0, 1] \cup [2, 3]$. Take $x = 2$.) In (11) of AI (2006), if " $h(u, 0)$ " is on the boundary of the support of continuously distributed $F_{y_i^0|00}$, then the argument used in their proof fails in general. Since h is strictly monotonic, $h(h^{-1}(y; 1), 0)$ being a boundary point in the support of a continuous distribution $\mathbb{Y}_{00}$ corresponds to y being in a probability measure zero set in $\mathbb{Y}_{01}$. Hence the quantifier "for almost every $y \in \mathbb{Y}_{11}$ ".

The map $\Delta(u)$ in (5) can be interpreted as a "shift function" studied by Doksum (1974).³ The within-group time invariance (WGTI) assumption of AI (2006), that is $u_i \perp\!\!\!\perp t_i | d_i$ implies (5) trivially with $\Delta(u) = 0$. The converse does not hold in general. In this sense, (5) is a weaker identifying assumption.

2.2.3. Discussion: Advantages and Disadvantages of the Proposed Method

We conclude this section by summarizing the comparison of two models, CIC and DDID. First, we contrast two models by comparing (4) and (5).⁴ To identify and estimate the whole counterfactual CDF, the two approaches assume the common time effect on CDFs in fundamentally different ways. (4) assumes that the proportion of treatment group units with unobservable below u changes similarly from period 0 to 1 in the treatment and in the control group, for all u . Instead (5) assumes that the difference between the unobservables of two units at the same rank of the unobservable's distribution at period 0 and 1 is the same in the treatment and in the control group. In short, while (4) imposes the common temporal changes in ranks of a given value u , (5) assumes the common rank-preserving temporal changes in u given a rank. A technical similarity here is that both models

³The shift function here cannot be interpreted as the "common quantile effect" in general. In fact, the common time effect on quantiles results in a different model. See sec. 3.3 of AI (2006) for details.

⁴Comparison of (2) and premises of Proposition 1 can be similarly done.

²With $\Delta(y) = h(h^{-1}(y; 1) + \Delta_U(h^{-1}(y; 1)); 0) - y$.

spend effectively one moment condition⁵ per identification of one functional value of the counterfactual CDF.⁶ In this regard, we cannot say one model assumption is more restrictive than the other.

Two models, however, have quite different properties due to the different nature of the common time effect. First, DDID is lack of structural interpretability. For instance, (5) holds if the period 1 unobservable is equal to the period 0 unobservable plus a constant time effect common to all units. It seems to be harder to find such a special case for (4), which can be disadvantage of the proposed method if a further structural interpretation is desired. Note that the original model proposed by AI (2006) can be taken as the leading example of structures that satisfy (5). Second, allowing covariates (e.g., pre-treatment characteristics) and discrete outcomes (e.g., count variables) is much easier with DDID. It is not impossible to allow them for the CIC model but it may require some extra care (e.g., sec. 4 of AI (2006) for discrete outcomes, Melly and Santangelo (2015, manuscript) for allowing covariates). For DDID, no special treatment for these cases is necessary, and the method proposed in this article remains valid. Third, the DDID assumption is partially testable via monotonicity of the counterfactual CDF while CIC does not imply any testable restriction. Note that the identifying expression in Proposition 1 is monotone increasing in y by construction. Fourth, the implied mean models differ in DDID and CIC. Under DDID, the usual DID assumption is implied for any scaling. In contrast, the CIC-implied mean model generally conflicts with the usual DID assumption. So, our method has an advantage when theoretical consistency with the usual DID assumption is important.

2.3. Characterization of DDID: Revisiting Roth and Sant'Anna (2023)

Assumption 1 requires a common time effect on the rank of each y . To properly guide its empirical plausibility, we revisit Roth and Sant'Anna (2023, Proposition 3.2; RS hereafter) and propose an alternative decomposition result. We argue that the interpretation by RS, which suggests that Assumption 1 holds “essentially only if” the population can be partitioned into two subgroups, one with randomly assigned treatment and the other one with time-stable untreated outcome distribution (Remark 2 of RS 2023), overlooks many cases and is too restrictive for general empirical guidance. First, we show that the (untreated) outcome density can always be decomposed into the time-stable part and the net temporal change part. Then, we prove that Assumption 1 holds if and only if the net temporal change of densities is common across groups; this does not necessitate randomly assigned treatment or time-stable untreated outcome distribution for any subgroup. Basically, the result shows that Assumption 1 imposes a common time effect on the Radon-Nikodym densities of the untreated potential outcomes.

2.3.1. An Alternative Decomposition

To present the result, we assume that the distribution of $y_i^0 | \mathbf{x}_i = \mathbf{x}, d_i = d, t_i = t$ for all $\mathbf{x} \in \text{spt}(\mathbf{x})$, $d = 0, 1$ and $t = 0, 1$ has a common dominating positive σ -finite measure, λ .⁷ A Radon-Nikodym density⁸ is denoted as $f_{\mathbf{x},dt}(y)$ with $f_{\mathbf{x},dt}(y) \geq 0$ for all $y \in \mathbb{R}$. Then, it is always possible to decompose each density as follows: for $y \in \mathbb{R}$, $\mathbf{x} \in \text{spt}(\mathbf{x})$, $d = 0, 1$, and $t = 0, 1$, we have

$$f_{\mathbf{x},dt}(y) = g_{\mathbf{x},dt}(y) + h_{\mathbf{x},d}(y) \quad (6)$$

where $g_{\mathbf{x},dt}(y) \equiv f_{\mathbf{x},dt}(y) - \min\{f_{\mathbf{x},d1}(y), f_{\mathbf{x},d0}(y)\}$ and $h_{\mathbf{x},d}(y) \equiv \min\{f_{\mathbf{x},d1}(y), f_{\mathbf{x},d0}(y)\}$. By construction, the equality holds with $g_{\mathbf{x},dt}(y) \geq 0$ and $h_{\mathbf{x},d}(y) \geq 0$ for each $(y, \mathbf{x}, d, t)$. Note again that the decomposition in (6) holds regardless of Assumption 1. It should not be interpreted as a “mixture of two groups” since $g_{\mathbf{x},dt}(y)$ only captures net changes in $f_{\mathbf{x},dt}(y)$. (This point can be further clarified by an example. See Appendix A in the supplementary materials.)

Before we present the decomposition result, we first give a natural interpretation of $g_{\mathbf{x},dt}(y)$: the net flow of density at y . For example, if $f_{\mathbf{x},d1}(y) > f_{\mathbf{x},d0}(y)$ at y , then the density increased at y over time [net inflow], and, by construction, we have $g_{\mathbf{x},d1}(y) = f_{\mathbf{x},d1}(y) - f_{\mathbf{x},d0}(y)$. If $f_{\mathbf{x},d1}(y) < f_{\mathbf{x},d0}(y)$, then the density decreased at y over time [net outflow], and we have $g_{\mathbf{x},d0}(y) = f_{\mathbf{x},d0}(y) - f_{\mathbf{x},d1}(y)$. Based on the interpretation, we give Definition 2 and the alternative decomposition result, Proposition 2.

Definition 2 (Net Flow of Density). In (6), we call $g_{\mathbf{x},d0}(y)$ the net outflow of density at y , and $g_{\mathbf{x},d1}(y)$ the net inflow of density at y (over the periods $t = 0$ and 1). The pair $(g_{\mathbf{x},d0}(y), g_{\mathbf{x},d1}(y))$ is called the net flow of density at y .

Remark 3 (In or out, pointwise). We have either $g_{\mathbf{x},d0}(y) = 0$ or $g_{\mathbf{x},d1}(y) = 0$ λ a.e..

Remark 4 (In and out, overall). The mass of the net inflow and the net outflow must be equal. that is $\int g_{\mathbf{x},d1}(y) d\lambda = \int g_{\mathbf{x},d0}(y) d\lambda$ since $1 = \int g_{\mathbf{x},dt}(y) d\lambda + \int h_{\mathbf{x},d}(y) d\lambda$ for $t = 0, 1$.

Proposition 2 (DDID). Assume the distribution of $y_i^0 | \mathbf{x}_i = \mathbf{x}, d_i = d, t_i = t$ for all $\mathbf{x} \in \text{spt}(\mathbf{x})$, $d = 0, 1$ and $t = 0, 1$ have a common dominating positive σ -finite measure, λ . Then, the net flow of density does not depend on d (λ a.e.) if and only if Assumption 1 holds.

Proof. “Only if” part is trivial. To show “if” part, suppose Assumption 1, which is equivalent to $f_{\mathbf{x},11}(y) - f_{\mathbf{x},10}(y) = f_{\mathbf{x},01}(y) - f_{\mathbf{x},00}(y)$ λ a.e.. Note that $g_{\mathbf{x},dt}(y) = f_{\mathbf{x},dt}(y) - [f_{\mathbf{x},d1}(y) - (f_{\mathbf{x},d1}(y) - f_{\mathbf{x},d0}(y))_+]$ where $(x)_+ = x1_{(x>0)}$, and that $g_{\mathbf{x},dt}(y)$ depends only on $f_{\mathbf{x},d1}(y) - f_{\mathbf{x},d0}(y)$ for each $t = 0, 1$. Hence the result. $\square$

⁷In Roth and Sant'Anna (2023), covariates are not explicitly allowed but the arguments in their proofs remain valid after conditioning on covariates. If $\mathbf{x}$ is chosen to be a constant, then we are back to the original setting with no covariates. Note that Assumption 1 with nontrivial covariates $\mathbf{x}$ does not generally imply the unconditional version (without any $\mathbf{x}$).

⁸For example, this can be a probability density function (with Lebesgue measure) and a probability mass function (with the counting measure) in the case of a continuous and discrete outcome variable, respectively.

⁵A conditional moment for each value of covariates if we allow covariates in both models.

⁶Strictly speaking, the CIC model in its original form proposes WGTI assumption which is stronger than necessary for identification.

Since the net flow of density can be seen as the time effect on the density, Proposition 2 shows that the decomposition result is just another way to express the common time effect assumption. In particular, it does not necessitate randomly assigned treatment or time-stable untreated outcome distribution for any subgroup.

To clarify the last point, we may compare the RS characterization and our interpretation in the context of structural assumptions in Section 2.2.1. (For simplicity, omit $\mathbf{x}$.) RS interpretation can be stated as follows: in both groups, there is a partition into two subpopulations such that one subpopulation did not experience any intertemporal changes in its distribution of u , and the other experienced the common intertemporal changes in the distribution of u across groups. What we claim here is that this is only a special case. We show that DDID holds if and only if the net intertemporal changes in the distribution of u are the same across groups regardless of who experienced what changes in u .

2.4. Multilevel Treatment

The identification result is extended to the case with multilevel treatment. Suppose there are V different treatment levels: $v = 1, \dots, V$. For each treatment level v , define a dummy variable w_i^v that takes value 1 if individual i is in the treated group and the level of treatment received is v , and value 0 otherwise. Then, Assumption 1 can be modified to the following: for each $v = 1, \dots, V$, for all $y \in \mathbb{R}$ and $\mathbf{x} \in \text{spt}(\mathbf{x}_i)$,

$$F_{y_i^0|\mathbf{x},11v}(y) - F_{y_i^0|\mathbf{x},10v}(y) = F_{y_i^0|\mathbf{x},01}(y) - F_{y_i^0|\mathbf{x},00}(y) \quad (7)$$

where $F_{y_i^0|\mathbf{x},d1v}(y) \equiv P(y_i^0 \leq y | \mathbf{x}_i = \mathbf{x}, d_i = d, t_i = t, w_i^v = 1)$. Under the modified distributional DID assumption (7) together with the correspondingly modified common support assumption, it can be shown similarly as in Theorem 1 that the following observable moment condition yields identification of $F_{y_i^0|11}(y)$ for $y \in \mathbb{R}$.

$$F_{y_i^0|11}(y) = E \left[\sum_{v=1}^V (p_{11v}(\mathbf{x}_i) / p_{11}) \{ F_{y_i^0|\mathbf{x}_i,10v}(y) + F_{y_i^0|\mathbf{x}_i,01}(y) - F_{y_i^0|\mathbf{x}_i,00}(y) \} \right] \quad (8)$$

where $p_{11v}(\mathbf{x}_i) \equiv P(d_i t_i w_i^v = 1 | \mathbf{x}_i)$. Compared to (3), the moment condition (8) has separate terms from different treatment levels. It is discussed how (8) motivates a two-stage estimation procedure in Section 3.1.

3. Inference

We propose DF-based methods. QTT is estimated by taking the difference of the left-inverted status quo and counterfactual DF estimators. The left-inverted DFs serve as QF estimators. To construct confidence bands for QFs, we first compute confidence bands for DFs by the bootstrap method, and left-invert the DF-bands. The Minkowski difference of QF-bands delivers the QTT band. In this section, a simple two-stage estimator of DFs and its asymptotic properties are the main building blocks. Properties of QF and QTT estimators are derived as corollaries.

3.1. Estimation of DFs, QFs, and QTT

To estimate counterfactual DF, a two-stage procedure is motivated by reformulating (3) (or (8) in the case of multilevel treatment). For notational convenience, define $1_{i,dt} \equiv 1_{(d_i=d \& t_i=t)}$. Also, write $F(\cdot) \equiv (F_1(\cdot), F_2(\cdot))' \equiv (F_{y_i^1|11}(\cdot), F_{y_i^0|11}(\cdot))'$. In the binary treatment case, by noting that $F_{y_i|\mathbf{x}_i,dt}(y) = E[(1_{i,dt}/p_{dt}(\mathbf{x}_i))1_{(y_i \leq y)} | \mathbf{x}_i]$ for each (d, t) , we can show (3) implies a moment condition

$$E \left[\frac{p_{11}(\mathbf{x}_i)}{p_{11}} \left(\frac{1_{i,10}}{p_{10}(\mathbf{x}_i)} + \frac{1_{i,01}}{p_{01}(\mathbf{x}_i)} - \frac{1_{i,00}}{p_{00}(\mathbf{x}_i)} \right) 1_{(y_i \leq y)} \right] - F_{y_i^0|11}(y) = 0. \quad (9)$$

Similarly, in the case of V -multilevel treatment, (8) implies

$$E \left[\sum_{v=1}^V \frac{p_{11v}(\mathbf{x}_i)}{p_{11}} \left(\frac{1_{i,10} w_i^v}{p_{10v}(\mathbf{x}_i)} + \frac{1_{i,01}}{p_{01}(\mathbf{x}_i)} - \frac{1_{i,00}}{p_{00}(\mathbf{x}_i)} \right) 1_{(y_i \leq y)} \right] - F_{y_i^0|11}(y) = 0 \quad (10)$$

where $p_{d1v}(\mathbf{x}_i) \equiv P(d_i = d, t_i = t, w_i^v = 1 | \mathbf{x}_i)$. In the reformulated moment conditions, each conditional cumulative distribution function is now replaced with a dummy variable divided by a propensity score. In (10), the weight multiplied to $1_{(y_i \leq y)}$ consists of V terms each of which corresponds to a treated subgroup with a different treatment level. When $V = 1$, (10) collapses to (9). For simplicity, we focus on the binary treatment case (9).

The proposed two-stage procedure is as follows. (Stage 1) Compute an estimate $\hat{p}_{dt}(\mathbf{x})$ for each (d, t) by running a non-parametric multinomial regression. (Stage 2) Solve the sample analog of (9) where $\hat{p}_{dt}(\mathbf{x})$'s are substituted for $p_{dt}(\mathbf{x})$'s. In the first stage, propensity scores, $p_{dt}(\mathbf{x})$'s, can be estimated non-parametrically using the multinomial logit series estimator (MLSE) of Cattaneo (2010). MLSE approximates the log-odds ratios

$$\log \left(\frac{p_{dt}(\mathbf{x})}{p_{11}(\mathbf{x})} \right) = R_K(\mathbf{x})' \gamma_{K,dt} + \text{error} \quad (11)$$

for $(d, t) = (0, 0), (0, 1), (1, 0)$ where $R_K(\mathbf{x})$ is a vector of K basis terms such as polynomials and splines. The coefficients $\gamma_K = (\gamma'_{K,00}, \gamma'_{K,01}, \gamma'_{K,10})'$ can be estimated by solving a quasi-maximum likelihood problem

$$\hat{\gamma}_K = \arg \min_{\gamma_K} \frac{1}{n} \sum_{i=1}^n \sum_{(d,t)} \log \left\{ \frac{\exp(R_K(\mathbf{x}_i)' \gamma_{K,dt})}{\sum_{(d,t)} \exp(R_K(\mathbf{x}_i)' \gamma_{K,dt})} \right\}^{1_{i,dt}} \quad (12)$$

where $\gamma_{K,11} = 0$ is imposed. Then, propensity scores are computed as

$$\hat{p}_{dt}(\mathbf{x}) = \frac{\exp(R_K(\mathbf{x})' \hat{\gamma}_{K,dt})}{\sum_{(d,t)} \exp(R_K(\mathbf{x})' \hat{\gamma}_{K,dt})} \quad (13)$$

for $(d, t) = (0, 0), (0, 1), (1, 0), (1, 1)$. Note that we have $p_{11}(\cdot) = 1 - \sum_{(d,t) \neq (1,1)} p_{dt}(\cdot)$ by construction.

Similarly, to estimate the status quo distribution function, in the first stage, p_{11} is estimated by computing the sample average

of $1_{i,11}$. In the second stage, we can solve for $F_{y_i|11}(y)$ from the sample analogue of the following moment condition

$$E \left[\frac{1_{i,11}}{p_{11}} 1_{(y_i \leq y)} \right] - F_{y_i|11}(y) = 0 \quad (14)$$

with the estimated p_{11} plugged in.

In short, the two-stage estimator of DF, $\hat{F}(\cdot) \equiv (\hat{F}_1(\cdot), \hat{F}_2(\cdot))'$, is defined to be the unique solution to the sample moment conditions (14) and (9) with the first stage estimate of $\mathbf{p}(\cdot) \equiv (p_{11}, p_{00}(\cdot), p_{01}(\cdot), p_{10}(\cdot))'$ substituted. Then, QF and QTT estimators are computed as its left-inverse and the difference of the inverses that is $\hat{F}^{\leftarrow}(\cdot) \equiv (\hat{F}_1^{\leftarrow}(\cdot), \hat{F}_2^{\leftarrow}(\cdot))'$ and $\hat{\Delta}(\cdot) \equiv \hat{F}_1^{\leftarrow}(\cdot) - \hat{F}_2^{\leftarrow}(\cdot)$, respectively.

3.2. Asymptotic Properties of Estimators

The uniform consistency and weak convergence of DF estimator $\hat{F}(\cdot)$ are established. Based on the theoretical framework of the generic Z-estimator (Galvao and Wang 2015), the first-stage nonparametric estimation and the infinite-dimensionality of the moment condition are accounted for. To show uniform consistency of $\hat{F}(\cdot)$, the following regularity conditions are assumed.

Assumption 6. (a) For all j , $p_j(\cdot)$ is s times continuously differentiable with $s/d_x > 2\eta + 1$ where $\eta = 1$ (power series) or $\eta = \frac{1}{2}$ (spline). The derivatives are uniformly bounded. (b) $\mathbf{x}$ is continuously distributed with density bounded and bounded away from zero on its compact support. (c) $K = n^\nu$ with $2s/d_x > 1/\nu > 4\eta + 2$ where $\eta = 1$ (power series) or $\eta = 1/2$ (spline).

Assumption 6 is a set of primitive conditions under which the first-stage estimator of $\mathbf{p}(\cdot)$ converges at a faster rate than $n^{-1/4}$. The assumptions are similar to Assumption 8 and the premises in Theorem 8 of Cattaneo (2010).⁹ The smoothness requirement for the spline case is slightly strengthened here, and it is in fact for the weak convergence, not for uniform consistency. We need some notations to present the consistency result. Define $\mathcal{L} \equiv l^\infty(\mathcal{Y}) \times l^\infty(\mathcal{Y})$ and $\|g\|_{\mathcal{L}} \equiv \sup_{y \in \mathcal{Y}} |g_1(y)| + \sup_{y \in \mathcal{Y}} |g_2(y)|$

for $g = (g_1, g_2)' \in \mathcal{L}$. To denote outer probability measure and outer expectation, “*” is used as a superscript. $F_o = (F_{o,1}, F_{o,2})'$ denotes the true DFs.

Theorem 2 (Uniform Consistency of DF Estimator). Suppose Assumptions 1–3 and 6. Then, $\|\hat{F}(\cdot) - F_o(\cdot)\|_{\mathcal{L}} = o_{p^*}(1)$ (in outer probability)

Since uniform convergence of $\hat{F}(\cdot)$ implies pointwise convergence, under the same assumptions of Theorem 2, pointwise convergence (weak consistency) of the QF estimator follows: $\hat{F}^{\leftarrow}$ converges in probability to $F_o^{\leftarrow}$ at continuous points of $F_o^{\leftarrow}$ (Lemma 21.2, der Vaart 2000). With mild conditions added, we can show (almost) uniform consistency of the QF estimator. Suppose that the support of $F_{o,k}(\cdot)$ [or $\mathcal{Y} = [a, b]$] is compact in $\mathbb{R}$ and $F_{o,k}^{\leftarrow}(\cdot)$ has a finite number of jump points in $[0, 1]$

[or $[\underline{\tau}_k, \bar{\tau}_k]$ where $\underline{\tau}_k \equiv F_{o,k}(a)$ and $\bar{\tau}_k \equiv F_{o,k}(b)$] for $k = 1, 2$. Then, consider a set constructed by subtracting arbitrarily small ε neighborhoods around the jump points from $[0, 1]$ [or $[\underline{\tau}_k, \bar{\tau}_k]$]. Denote such set as $\mathcal{T}_{\varepsilon,k}$. A precise definition of $\mathcal{T}_{\varepsilon,k}$ is given in the proof. Let $\mathcal{T}_{\varepsilon}^J \equiv \mathcal{T}_{\varepsilon,1} \cap \mathcal{T}_{\varepsilon,2}$. The following corollary shows uniform consistency of $\hat{F}^{\leftarrow}$ in $l^\infty(\mathcal{T}_{\varepsilon,1}) \times l^\infty(\mathcal{T}_{\varepsilon,2})$ and that of $\hat{\Delta}$ in $\mathcal{T}_{\varepsilon}^J$. Note that $\mathcal{T}_{\varepsilon}^J = [0, 1]$ [or $[\underline{\tau}_1, \bar{\tau}_1] \cap [\underline{\tau}_2, \bar{\tau}_2]$] if there is no jump point.

Corollary 2 ((Almost) Uniform Consistency of QF and QTT Estimators). Suppose that the assumptions of Theorem 2 hold. Assume that (a) the support of the outcome is compact in $\mathbb{R}$, and (b) $F_{o,k}^{\leftarrow}(\cdot)$ has a finite number of jump points in $[0, 1]$ for $k = 1, 2$. Then, (i) $\|\hat{F}^{\leftarrow}(\cdot) - F_o^{\leftarrow}(\cdot)\|_{l^\infty(\mathcal{T}_{\varepsilon,1}) \times l^\infty(\mathcal{T}_{\varepsilon,2})} = o_{p^*}(1)$ and (ii) $\|\hat{\Delta}(\cdot) - \Delta_o(\cdot)\|_{l^\infty(\mathcal{T}_{\varepsilon}^J)} = o_{p^*}(1)$. In particular, if $F_o^{\leftarrow}(\cdot)$ is continuous, then the convergence in (i) and (ii) hold in $l^\infty[0, 1] \times l^\infty[0, 1]$ and $l^\infty[0, 1]$, respectively. Alternatively, if (a)' $\mathcal{Y} = [a, b]$ is compact in $\mathbb{R}$, and (b)' $F_{o,k}^{\leftarrow}(\cdot)$ has a finite number of jump points in $(\underline{\tau}_k, \bar{\tau}_k)$ for $k = 1, 2$ where neither $\underline{\tau}_k$ nor $\bar{\tau}_k$ is a jump point, then (i) and (ii) hold.

The additional conditions imposed in Corollary 2 are mild. The compactness of the support is standard in the literature. A component of $F_o^{\leftarrow}(\cdot)$ having finite jump points is equivalent to the condition that the corresponding distribution has a finite number of intervals in $\mathbb{R}$ with probability measure zero. We rule out the possibility of infinite jump points here since they can be dense in $[0, 1]$ (e.g., Cantor-Lebesgue function). Still, these conditions allow generic outcomes in practice such as a discrete variable with a finite support and a bounded mixture variable with a finite number of measure-zero intervals in $\mathbb{R}$. Interestingly, uniform consistency is achieved allowing the outcome variable to have arbitrary probability mass points (note: mass points do not create jump discontinuity in $F_o^{\leftarrow}(\cdot)$).

For weak convergence of $\hat{F}(\cdot)$, we additionally assume the following regularity condition.

Assumption 7. $F_{y_i^0|\mathbf{x}_i,dt}(q)$ is s times continuously differentiable in $\mathbf{x}$ with $s/d_x > 2\eta + 1$ for $d = 0, 1$ and $t = 0, 1$. The derivatives are uniformly bounded in $(y, \mathbf{x})$.

Assumption 7 is needed for uniform linearizability of $\hat{F}(\cdot)$. Similar conditions are used in the related literature (e.g., Rothe 2010; Chernozhukov, Fernández-Val, and Melly 2013; Hsu, Lai, and Lieli 2020). Next, we introduce some notations to denote the influence function. Recall that $\mathbf{F} = (F_1, F_2)'$. We can write the moment functions in (14) and (9) as $g_{i,1}(F_1, \tilde{\mathbf{p}}, y)$ and $g_{i,2}(F_2, \tilde{\mathbf{p}}, y)$, respectively so that $E[g_i(F_o, \mathbf{p}_o, y)] = 0$ with $g_i = (g_{i,1}, g_{i,2})'$. Define $G_{p_{11},y} \equiv E[\partial g_i(F_o, \mathbf{p}_o, y)/\partial p_{11}]$, and $G_{p_{-11},y} \equiv E[\partial g_i(F_o, \mathbf{p}_o, y)/\partial \mathbf{p}_{-11}(\mathbf{x}_i)|\mathbf{x}_i]$. Then, the influence function can be written as $\tilde{\psi}(y) \equiv g_i(F_o, \mathbf{p}_o, y) + g_i^c(F_o, \mathbf{p}_o, y)$ where $g_i^c(F_o, \mathbf{p}_o, y) \equiv G_{p_{11},y}(1_{i,11} - p_{o,11}) + G_{p_{-11},y}(\mathbf{D}_{i,-11} - \mathbf{p}_{o,-11}(\mathbf{x}_i))$ and $\mathbf{D}_{i,-11} = (1_{i,00}, 1_{i,01}, 1_{i,10})'$. The correction term $g_i^c(F_o, \mathbf{p}_o, y)$ is one of the kind discussed by Newey (1994). It reflects the influence of the first-stage nonparametric estimation on the estimation of DFs. Similar expressions were previously

⁹As discussed in Cattaneo (2010), at the cost of some additional notation, we can allow some components of $\mathbf{x}$ to be discrete variables with finite support points.

(derived in different contexts (e.g., Hirano, Imbens, and Ridder 2003; Abadie 2005; Firpo (2007); Cattaneo 2010).

Theorem 3 (Weak Convergence of DF Estimator). Suppose Assumption 1–3, 6, and 7. Then, $\sqrt{n}(\hat{F}(\cdot) - F_o(\cdot)) \rightsquigarrow \mathbb{G}(\cdot)$ in $\mathcal{L}$ where $\mathbb{G}(\cdot)$ is a two-dimensional zero mean Gaussian process with covariance function $E[\mathbb{G}(y_1)\mathbb{G}(y_2)'] = E[\tilde{\psi}(y_1)\tilde{\psi}(y_2)']$.

For the validity of the DF-band-inversion method of CFMW (2020), Theorem 3 and the bootstrap consistency for DF estimator (Theorem 5 in Section 3.3) are sufficient. For completeness of theory, under standard regularity conditions, we show weak convergence of the proposed QF and QTT estimators. Define their densities as $f_1(\cdot) \equiv F'_{o,1}(\cdot)$ and $f_2(\cdot) \equiv F'_{o,2}(\cdot)$ (f_k is continuously extended on the boundary points.)

Corollary 3 (Weak Convergence of QF and QTT Estimators). Suppose that the assumptions of Theorem 3 hold. Let $\mathbb{G} = (\mathbb{G}_1, \mathbb{G}_2)'$ be the zero mean Gaussian process in Theorem 3. Assume (a) the supports are compact intervals, and (b) $f_k(\cdot)$ is continuous and positive on the support for $k = 1, 2$. Then, (i) $\sqrt{n}(\hat{F}^{\leftarrow}(\cdot) - F_o^{\leftarrow}(\cdot)) \rightsquigarrow (\mathbb{H}_1(\cdot), \mathbb{H}_2(\cdot))'$ in $l^\infty(0, 1) \times l^\infty(0, 1)$ and (ii) $\sqrt{n}(\hat{\Delta}(\cdot) - \Delta(\cdot)) \rightsquigarrow \mathbb{H}_1(\cdot) - \mathbb{H}_2(\cdot)$ in $l^\infty(0, 1)$ where $\mathbb{H}_k(\cdot) \equiv -\mathbb{G}_k(F_{o,k}^{-1}(\cdot))/f_k(F_{o,k}^{-1}(\cdot))$ for $k = 1, 2$. Alternatively, if (a)' $\mathcal{Y} = [a, b]$ is compact in $\mathbb{R}$, and (b)' $f_k(\cdot)$ is continuous and positive on $\mathcal{Y}$, then (i) holds in $l^\infty[p_1, q_1] \times l^\infty[p_2, q_2]$ and (ii) holds in $l^\infty[p_1 \vee p_2, q_1 \wedge q_2]$ where $[p_k, q_k]$ is an arbitrary subinterval of $(F_{o,k}(a), F_{o,k}(b))$ for $k = 1, 2$ such that $p_1 \vee p_2 < q_1 \wedge q_2$. (Such (p_k, q_k) may not exist if $\mathcal{Y}$ is too narrow.)

The proposed bootstrap procedure in this article is a DF-based method and it does not rely on Corollaries 2 or 3 at all. The motivation for DF-based method is that the conditions of Corollary 3 are often too restrictive in practice. For example, discrete outcome variables are not allowed. Still, these results are informative about the asymptotic properties of the proposed QF and QTT estimators when the standard regularity conditions are met.

In Theorem 4, the semiparametric efficiency bound for QTT is derived. Newey (1990) and Bickel et al. (1993) provide the theoretical framework. The proof is similar to those of Hahn (1998), Hirano, Imbens, and Ridder (2003), Firpo (2007), and Cattaneo (2010). The bound coincides with the pointwise asymptotic variance of the proposed estimator of QTT derived in Corollary 3. Recall the notation for the influence function, $\tilde{\psi}(\cdot) = (\tilde{\psi}_1(\cdot), \tilde{\psi}_2(\cdot))'$. Define $\psi^*(\tau) = -\tilde{\psi}_1(F_{o,1}^{-1}(\tau))/f_1(F_{o,1}^{-1}(\tau)) + \tilde{\psi}_2(F_{o,2}^{-1}(\tau))/f_2(F_{o,2}^{-1}(\tau))$ for $\tau \in (0, 1)$.

Theorem 4 (Semiparametric Efficiency). Suppose that the assumptions of Corollary 3 hold. Then, the semiparametric efficiency bound for $\Delta(\tau)$ is $E[\psi^*(\tau)^2]$.

3.3. Bootstrap Procedure

Weak consistency of the (nonparametric) bootstrap estimator is established and an algorithm for constructing simulta-

neous confidence bands is discussed. Suppose bootstrap data $\{(y_{b,i}, \mathbf{x}_{b,i}, d_{b,i}, t_{b,i})\}_{i=1}^n$ is drawn with replacement from $\{y_i, \mathbf{x}_i, d_i, t_i\}_{i=1}^n$. Note that the size of a bootstrap sample is assumed to be the same as that of the original sample. Denote bootstrap estimates as $\hat{F}_b(\cdot) = (\hat{F}_{b,1}(\cdot), \hat{F}_{b,2}(\cdot))'$.

Theorem 5 (Weak Consistency of Bootstrap DF Estimator). Suppose that the assumptions of Theorem 3 hold. Then, conditional on data, $\sqrt{n}(\hat{F}_b(\cdot) - \hat{F}(\cdot)) \rightsquigarrow \mathbb{G}(\cdot)$ in $l^\infty(\mathcal{Y}) \times l^\infty(\mathcal{Y})$.

Theorem 5 together with 3 validates the proposed bootstrap procedure below (Algorithm 1 and Corollary 3 of CFMW 2020): under our model assumptions, the proposed confidence bands for DFs, QFs and QTT cover the true functions simultaneously with the pre-specified probability. Neither compactness of $\mathcal{Y}$ (or the support) nor well-behavedness of densities is assumed here. In Corollary 3, these assumptions are needed for the Hadamard differentiability of the left-inverse operator so that the functional delta method can deliver the limiting process of QF estimators. The proposed bootstrap method avoids the use of the functional delta method by applying left-inversion to bootstrap-generated DF-bands. As a result, the pre-specified simultaneous coverage rate of confidence bands is achieved under much less restrictive conditions. The procedure is valid for discrete, continuous, and mixed outcome variables while *any* alternative method based on Corollary 3 (e.g., multiplier bootstrap) would break down, when the outcome is discrete, for example.

To present the bootstrap algorithm for simultaneous confidence bands, we need some notations. Let T be a subset of $\mathcal{Y}$. In practice, we choose some T , and $F_o(\cdot)$ is estimated at each point of T . As discussed in CFMW (2020), T can be the support itself when y is a discrete variable with a finite support. Generally, T can be chosen as a set of grid points in the region of interest $\mathcal{Y}$. We denote the chosen set for F_k as T_k when they are allowed to be different. The shaping operator $\mathcal{S}$ is defined as $\mathcal{S}(f) \equiv \mathcal{M}(0 \vee f \wedge 1)$ for the given rearrangement operator $\mathcal{M}$ where the maximum and minimum are taken pointwise. $\mathcal{M}(f)$ is constructed by one-to-one mapping each element in T to an element in the range of f in increasing order. To see this in detail, suppose T has L points, $y_1 < \dots < y_L$, and the domain of f is T . We can denote the range of f as a family $\{f(y_l) : l = 1, \dots, L\}$. Then, we can reindex the elements of the family as $\tilde{f}_l$ for $l = 1, \dots, L$ such that $\tilde{f}_l \leq \tilde{f}_{l'}$ if $l \leq l'$. Such reindexing is possible since the family is finite. The rearranged map $\mathcal{M}(f)$ is defined as $\mathcal{M}(f)(y_l) \equiv \tilde{f}_l$ for $l = 1, \dots, L$. For properties of $\mathcal{S}$, see Section A of the supplementary appendix of CFMW (2020). The difference of two bands in $l^\infty(0, 1)$, $V(\cdot) = [v_1(\cdot), v_2(\cdot)]$ and $U = [u_1(\cdot), u_2(\cdot)]$ are defined with pointwise Minkowski difference, that is $(V - U)(\tau) \equiv [v_1(\tau) - u_2(\tau), v_2(\tau) - u_1(\tau)]$ for $\tau \in (0, 1)$.

Algorithm 1 (Bootstrap Algorithm for Simultaneous Confidence Bands; CFMW 2020). 1. Obtain bootstrap draws of the estimator $\hat{F}_b^{(j)}(\cdot) = (\hat{F}_{b,1}^{(j)}(\cdot), \hat{F}_{b,2}^{(j)}(\cdot))'$ $j = 1, \dots, B$ where B is the number of bootstrap draws.

2. For each $y \in T_k$ and $k = 1, 2$, compute the robust standard error of $\hat{F}_k(\cdot)$: $\hat{s}_k(y) = (\hat{Q}_k(0.75, y) - \hat{Q}_k(0.25, y)) / (\Phi^{-1}(0.75) - \Phi^{-1}(0.25))$ where $\hat{Q}_k(\alpha, y)$ denotes the empirical α -quantile

of the bootstrap sample $\{\hat{F}_{b,k}^{(j)}(y)\}_{j=1}^B$, and Φ^{-1} denotes the inverse of the standard normal distribution.

3. Compute the critical value $c(p) = p$ -quantile of $\left\{ \max_{y \in T, k=1,2} \left| \hat{F}_{b,1}^{(j)}(y) - \hat{F}_1(y) \right| / \hat{s}_k(y) \right\}_{j=1}^B$.

4. Construct preliminary joint DF-bands of level p as $[L'_k(y), U'_k(y)] = [\hat{F}_k(y) \pm c(p) \hat{s}_k(y)]$, $y \in T$, $k = 1, 2$. For each $k = 1, 2$, impose the shape restriction on $\hat{F}_k(\cdot)$, $L'_k(y)$ and $U'_k(y)$ that is $L_k \equiv \mathcal{S}(L'_k)$ and $U_k \equiv \mathcal{S}(U'_k)$.

5. Report $I_k = [L_k, U_k]$ for $k = 1, 2$ as p -level simultaneous confidence bands for DFs. Report $I_k^{\leftarrow} = [L_k^{\leftarrow}, U_k^{\leftarrow}]$ or the support-restricted version $\tilde{I}_k^{\leftarrow} = I_k^{\leftarrow} \cap T_k$ for $k = 1, 2$ as p -level simultaneous confidence bands for QFs.

6. Report $I_{\Delta}^{\leftarrow} \equiv I_1^{\leftarrow} - I_2^{\leftarrow}$ or the support-restricted version $\tilde{I}_{\Delta}^{\leftarrow} \equiv \tilde{I}_1^{\leftarrow} - \tilde{I}_2^{\leftarrow}$ as p -level simultaneous confidence bands for QTT.

3.4. Testing DDID Assumption: Chernozhukov-Chetverikov-Kato (CCK) Test

To assess the credibility of Assumption 1, we may consider testing the monotonicity presented in Corollary 1 in Section 2.1. RS (2023) first introduced the idea of testing monotonicity via a set of moment inequalities. We take a similar approach, but our method is different from the RS method in three aspects. First, to flexibly apply the method to outcome variables with infinitely many support points, we consider a procedure valid for a diverging number of moment inequalities. To this end, we apply the method proposed by Chernozhukov, Chetverikov, and Kato (2019, CCK). Second, the null hypothesis is formulated with moment inequalities that compare the counterfactual DF values over *all* pairs of points in the support (or the grid) rather than only the pairs of adjacent points. The additional moment inequalities do not make the null hypothesis stronger than the one used in RS method, but they can still improve the power of the test in finite samples. Third, the estimated variance used in the test statistic is computed using an analytical formula rather than bootstrap samples. The use of bootstrapped variance can lead to an undesirably high false rejection rate. Simulation evidence for the second and third points will be discussed in Section 4.

The setup of the testing problem is as follows. Suppose that the set T (discussed in Section 3.3) has L points, $y_1 < \dots < y_L$. We aim to test the monotonicity over points in T . We may instead use a set of grid points explicitly growing (as n tends to infinity). Note that the CCK method allows the number of inequalities to diverge at the order of $\exp(n^C)$ for some $C > 0$. For notational simplicity, define $\mu_{l,m} \equiv F_{y_l|11}(y_l) - F_{y_l|11}(y_m)$ for $l, m \in \{1, \dots, L\}$ with $l < m$. Then, the null hypothesis to be tested is $H_0 : \mu_{l,m} \leq 0$ for all $1 \leq l < m \leq L$ against the alternative $H_1 : \mu_{l,m} > 0$ for some (l, m) . The null hypothesis H_0 here is equivalent to that of RS method, which effectively imposes $m = l + 1$ in our H_0 . All inequalities with $m - l \geq 2$ are implied by the inequalities that compare consecutive points y_l and y_{l+1} for some l s.

The test statistic is computed as the maximum of studentized statistics for $\mu_{l,m}$ s:

$$\max_{1 \leq l < m \leq L} \frac{\sqrt{n} \hat{\mu}_{l,m}}{\hat{\sigma}_{l,m}} \quad (15)$$

where $\hat{\mu}_{l,m}$ is the estimated value of $\mu_{l,m}$ and its variance $\hat{\sigma}_{l,m}^2$ is computed based on the asymptotic variance formula $E[\tilde{\psi}_2(y_l) - \tilde{\psi}_2(y_m)]^2$ which is implied by Theorem 3. The decision rule is to reject the null when the test statistic exceeds a critical value. Since the test statistic above seeks the maximum of studentized decrease of the counterfactual DF over all arbitrary pairs of points (y_l, y_m) with $1 \leq l < m \leq L$, (given the same critical value,) the power of the test is not lower than the one that searches over pairs of consecutive points (y_l, y_{l+1}) with $l = 1, \dots, L - 1$. For example, the gain in power can be realized when the violation of monotonicity occurs with a larger magnitude over a larger region, that is, for some (y_l, y_m) with $m - l > 2$. We can intuitively interpret (15) as a way to formalize the following procedure: among all the pairs of points (y_l, y_m) in the support (or grids), we look for the pair that exhibits the largest violation of monotonicity (accounting for variance), and conduct a t -test on the monotonicity over the two points.

CCK proposes multiple methods for computing critical values to make the statistic to have size $\alpha \in (0, 1)$. Among them, we consider the self-normalized (SN) methods: the one-step SN critical value with size α is taken as

$$c^{SN}(\alpha) = \frac{\Phi^{-1}(1 - \alpha/L(L-1)/2)}{\sqrt{1 - \Phi^{-1}(1 - \alpha/L(L-1)/2)^2/n}}.$$

The two-step method may improve the power of the test by selecting nonbinding inequalities. Pick $0 < \beta_n < \alpha/2$ (e.g., $\beta_n = 0.001$). Define $\hat{L}_{SN} \equiv \{(l, m) : 1 \leq l < m \leq L, \sqrt{n} \hat{\mu}_{l,m} / \hat{\sigma}_{l,m} > -2c^{SN}(\beta_n)\}$. Denote the cardinality of $\hat{L}_{SN}$ as $|\hat{L}_{SN}|$. Then, the two-step SN critical value is

$$c^{SN,2S}(\alpha) = \begin{cases} \frac{\Phi^{-1}(1 - (\alpha - 2\beta_n)/|\hat{L}_{SN}|)}{\sqrt{1 - \Phi^{-1}(1 - (\alpha - 2\beta_n)/|\hat{L}_{SN}|)^2/n}} & \text{if } |\hat{L}_{SN}| \geq 1 \\ 0 & \text{if } |\hat{L}_{SN}| = 0. \end{cases}$$

4. .ado File Package for STATA and Monte Carlo Simulation

The STATA package for the proposed method is explained and some simulation results are discussed.

4.1. .ado File for DDID Estimator

The “ddid.ado” file (and “ddid0.ado”) that accompanies this article is briefly explained. Further details, including installation instructions, can be found in the “readme.txt” file. The ado program implements the method proposed in this article: it computes the DIDD estimator of QTT, its confidence band, and the monotonicity test. The program takes inputs of outcome variable y , group and time dummy variables, d and t , each of which takes 0 or 1, and a set of covariates $\mathbf{x}$. Here, $\mathbf{x}$ refers to the pre-selected regressors for the first-stage nonparametric multinomial logit regression. Note that $\mathbf{x}$ is different from the selected variables for the conditional joint probabilities related

to conditional cumulative distribution functions (at each grid point) that are required for monotonicity testing. The selected variables should be entered via global macros. If these macros are not entered, they will be filled with $\mathbf{x}$. It is recommended to enter the selected variables into the macros. Both model selection procedures should be done in separate routines beforehand. The ado program does not conduct any of these selections. Bootstrap is automatically performed to construct the confidence band for QTT. The grid points are either manually specified or taken as all realized values of the outcome variable. By default, 99 equispaced points between the realized minimum and maximum values of y are used.

4.2. Simulation Study

We consider two DGP designs in a class of histogram densities. The goal is to briefly check the performance of the proposed method for two simple setups, one with the DDID assumption satisfied and the other without it. Since any continuous distribution can be well-approximated by histogram densities with an increasing number of bins, our histogram densities with three bins can be straightforwardly extended to more realistic specifications if needed. Here, we may take the simple setup as a very crude approximation of a possible DGP, or even think of it as a part of a larger histogram. Technically, histogram densities are easy to handle as well. The corresponding CDF is a monotone increasing piecewise linear function. Its direct inversion will deliver the quantile function, which is also piecewise linear.

The setup of DGPs is as follows. Given a sample size n , half of the observations are from the first period $t = 0$ (with n set to be even). There is a single covariate x that follows a uniform distribution $U[0, 1]$. The treatment group membership is then drawn for each individual with $P(d_i = 1|x_i) = 0.5 + 0.1x$. The outcome variable y is drawn from histogram densities with three bins $[0, 1)$, $[1, 2)$, and $[2, 3]$. The constant probability density functions of y given (x, d, t) are shown in Table 1. Since each bin has a unit length, a density value can be interpreted

as the conditional probability of y being in each bin given (x, d, t) .

Note that Design 1 does not contradict the DDID assumption. The common time effect on the densities for the first, second, and third bin would be $-0.25 - 0.05x$, 0 , and $0.05 + 0.05x$, respectively. By some algebra, we can show that the true QTT at the 10th, 20th, ..., 90th percentiles are 0.83333, 0.9, 0.825, 0.75, 0.675, 0.6, 0.48214, 0.32143, and 0.16071, respectively. In contrast, DDID is violated under Design 2. The time effect on the $[1, 2)$ -density of control group is $-0.1 + 0.1x - s$ where $s \in (0, 0.5]$. If the same amount of change occurred in the treated group, the group's $[1, 2)$ -density would be $-s$, which is contradictory.

For estimation, we use 99 equispaced points between 0 and 3 as grid points. For estimating propensity scores and quantities related to the conditional CDF of y given (d, t, x) (at each grid point), the first k regressors were selected from 5th order polynomials of x by minimizing AIC. The number of bootstrap samples is 1000. We focus on the QTT at the 10th, 20th, ..., 90th percentiles. The results are given in Tables 2 and 3.

In Table 2, Bias and RMSE are relative to the true QTT values (i.e., the true QTT for each percentile is set to be 1). "CI" shows the coverage rate of the confidence interval at a given percentile or that of the joint confidence band. "SigTE" shows the rejection rate of the null hypothesis of zero QTT at a given percentile or all the percentiles considered. First, RMSE decreases across all percentiles as the sample size increases. The pointwise coverage rates of confidence intervals tend to increase, and they are all above 95% at $n = 1000$. The joint coverage rate increases as well, but it is still at 0.819 with $n = 1000$. The rejection rate of the significance test tends to increase, too. Since the true values of treatment effects at higher percentile are relatively smaller in magnitude, the speed of the rejection rate growth is slower at higher percentiles.

In Table 3, three types of CCK test were considered for testing monotonicity. For Design2, s takes 0.5, 0.2, and 0.05 representing the magnitude of violation (i.e., the negative $[1, 2)$ -density).

Table 1. DGPs—conditional PDF for (d, t) given x .

		$d = 0$			$d = 1$		
		$[0, 1)$	$[1, 2)$	$[2, 3]$	$[0, 1)$	$[1, 2)$	$[2, 3]$
Design1	$t = 0$	$0.3 + 0.05x$	$0.4 - 0.1x$	$0.3 + 0.05x$	$0.2 + 0.05x$	$0.55 - 0.1x$	$0.25 + 0.05x$
	$t = 1$	0.25	$0.4 - 0.1x$	$0.35 + 0.1x$	0	0.2	0.8
Design2 ($0 < s \leq 0.5$)	$t = 0$	$0.2 + 0.05x$	$0.6 - 0.1x$	$0.2 + 0.05x$	$0.45 + 0.05x$	$0.1 - 0.1x$	$0.45 + 0.05x$
	$t = 1$	$0.25 + 0.5s$	$0.5 - s$	$0.25 + 0.5s$	0	0.2	0.8

Table 2. QTT estimation (Design1).

n Percentile	250				500				1000			
	Bias	RMSE	CI	SigTE	Bias	RMSE	CI	SigTE	Bias	RMSE	CI	SigTE
10p	-0.021	0.474	0.974	0.504	0.010	0.371	0.969	0.825	-0.031	0.314	0.969	0.975
20p	-0.024	0.367	0.959	0.722	-0.010	0.260	0.97	0.955	-0.035	0.177	0.968	0.998
30p	-0.004	0.316	0.951	0.828	0.006	0.202	0.965	0.995	-0.007	0.143	0.963	1
40p	-0.018	0.336	0.96	0.782	0.002	0.223	0.972	0.991	-0.005	0.159	0.974	1
50p	-0.035	0.382	0.963	0.674	0.001	0.250	0.963	0.944	-0.005	0.176	0.969	1
60p	-0.043	0.435	0.96	0.498	-0.006	0.290	0.978	0.865	-0.012	0.198	0.966	0.996
70p	-0.022	0.559	0.949	0.333	0.008	0.386	0.959	0.71	-0.016	0.288	0.958	0.936
80p	0.054	0.794	0.938	0.206	0.071	0.564	0.963	0.403	-0.008	0.405	0.962	0.746
90p	0.474	20.455	0.596	0.454	0.128	0.894	0.939	0.191	0.007	0.612	0.96	0.355
Joint 10–90			0.521	0.967			0.809	0.999			0.819	1

Table 3. Rejection rate of monotonicity test.

Test	H_0	$\hat{\sigma}$	Stages	250		500		1000			
				Design1	Design2 ($s = 0.5$)	Design1	Design2 ($s = 0.5$)	Design1	Design2 ($s = 0.5$)	Design2 ($s = 0.2$)	Design2 ($s = 0.05$)
CCK1	All pairs	Analytic	One	0	0.977	0	0.999	0	1	0.554	0.014
CCK1	All pairs	Analytic	Two	0	0.974	0	0.999	0	1	0.558	0.014
CCK2	Adjacent only	Bootstrap	One	–	–	–	–	10.00	–	–	–
CCK2	Adjacent only	Bootstrap	Two	–	–	–	–	10.00	–	–	–
CCK3	Adjacent only	Analytic	One	–	–	–	–	–	0.029	0.002	0.002
CCK3	Adjacent only	Analytic	Two	–	–	–	–	–	0.029	0.002	0.002

CCK1 is the proposed method. Other tests, CCK2 and CCK3, make a good comparison to motivate the null hypothesis with all possible pairs of grid points and the use of the analytic formula for computing $\hat{\sigma}$. When $\hat{\sigma}$ was computed using bootstrap samples, the moment inequality test with pairs of adjacent points (CCK2) exhibited an undesirably high false rejection rate under Design1. When $\hat{\sigma}$ was computed based on the analytic formula, the test with inequalities on pairs of adjacent points (CCK3) had very low power. We also considered the bootstrapped critical value of CCK (the one-step version), but they tended to be larger than the analytic ones in our setup. So, it did not help improve the power of the test. On the contrary, CCK1 performs surprisingly well. It exhibited essentially no false rejection under Design 1, and very high power under Design 2 (with $s = .5$). This drastic improvement over CCK3 is mainly because the violation occurs with a relatively larger magnitude (for some (y_l, y_m) with $m - l > 2$) given the sample size, and CCK3 only checks violations between consecutive points and neglects all other possibilities. Note that when the violation magnitude s diminishes, the power of CCK1 also decreases. If violation only occurs between two adjacent points, the performance of CCK1 may not be better than CCK3 since the critical values tend to increase as the number of inequalities grows. Although monotonicity is violated over adjacent grid points that lie within the interval $[1, 2]$, CCK3 essentially fails to capture them here. These results show that the power of the test depends on the scale of violation and that the proposed method CCK1 can perform better than other methods considered when the violation occurs on a large scale. It would be interesting to study the possibility of refining the critical values for CCK3 (and CCK1) to improve its power for small scale violations between adjacent grid points. Note that the range of CCK3 test statistics under Design2 at $n = 1000$ and $s = 0.5$ was $[2.224, 3.697]$ and the critical values were about 3.3.

5. Concluding Remarks

We propose a difference-in-differences estimator of QTT. Compared to the existing models, the new DID assumption implies the usual DID assumption with covariates and the condition is testable via an implied restriction. We establish uniform consistency and weak convergence of the proposed estimator. The inference procedure is valid for generic outcomes including discrete variables.

For future research, there can be many interesting possibilities to pursue. For example, the model and method can be extended to allow multiple periods and different timing of the

treatment (e.g., Callaway and Sant'Anna 2020). Under such a setting, it would be possible to conduct a specification test (as in event studies) making use of multiple time points in the pre-period. The topic is currently being studied in a separate paper by an author. Another interesting direction would be on the relaxation of the distributional DID assumption. To identify QTT, the identifying condition is typically stronger than those needed in the case of ATT. Although our model provides a way to test the model assumption, it is still desirable to widen the set of admissible structures. For instance, we may take significant temporal changes in the conditional group propensity score (for some $\mathbf{x}$ values) as indirect evidence of selection bias and use set-identified counterfactual DFs for such $\mathbf{x}$ values.

Supplementary Materials

The supplementary materials include an illustrative example for Proposition 2, an empirical application of the proposed method, proofs, and Figure 2.

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